

Eigenvalues and Eigenvectors

An eigenvalue is a scalar λ such that $Av = \lambda v$ for a matrix A and vector v .

Eigenvectors are non-zero vectors that only change by a scalar factor when transformed.

The characteristic equation $\det(A - \lambda I) = 0$ is used to find eigenvalues.

Principal Component Analysis uses eigenvalue decomposition to reduce dimensionality.

Matrix diagonalization relies on finding linearly independent eigenvectors.